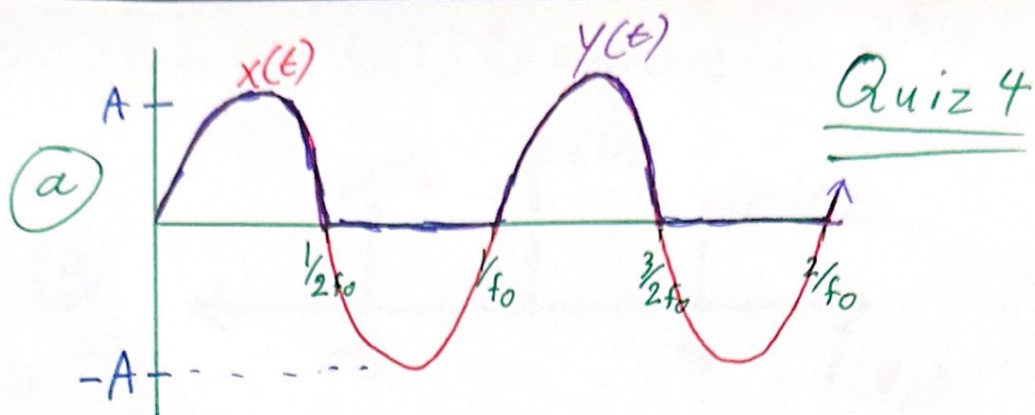


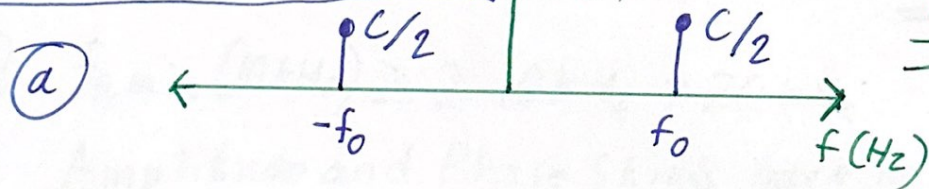


(b) Periodic. Period = $1/f_0$

(c) $Y(k) = f_0 \int_0^{1/2f_0} x(t) e^{-j2\pi kt f_0} dt$

(d) $Y(0) = 0$? is the DC component = 0?
 No. Signal is primarily in Positive Region exclusively

$g(t) = C \cdot \sin(2\pi f_0 t + \theta)$



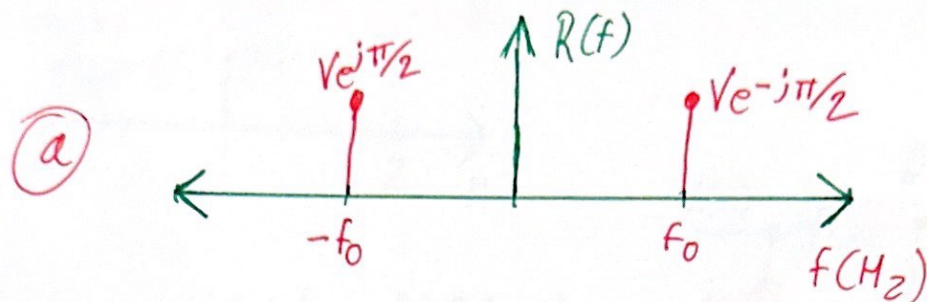
(b) Min. Sampling rate = $2 \cdot f_0$ as per Nyquist-Shannon Theorem

(c) Parg of Sinusoid = $\frac{\text{Amplitude}^2}{2}$

$\frac{2^2}{2} = \boxed{2 = P_{\text{arg}}}$ if $\theta = \frac{\pi}{2}$, $P=2$, since

Power is not affected by Phase Shift

$$r(t) = 2V_s \sin(2\pi \cdot f_0 t) \mid \text{Remember: } -j = e^{-j\pi/2} \mid \underline{\text{Quiz 6}}$$



(b) Min. Sampling Rate = $2 \times f_0$, as per Nyquist-Shannon Theorem

(c) $P_{\text{AVG. Sinusoid}} = \frac{A^2}{2} = \frac{(2.5)^2}{2} = \boxed{50 \text{ W}}$

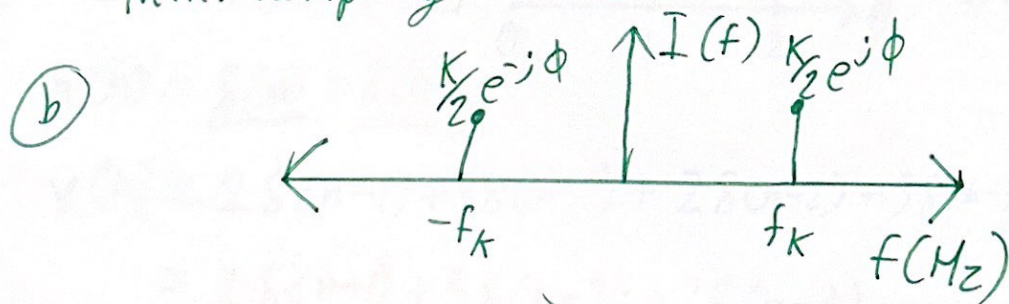
Frequency has No effect

$$i(t) = K \cdot \cos(2\pi f_k t + \phi)$$

Quiz 7

(a) $f_{s, \text{min}}(10 \text{ kHz}) \geq 2 \cdot 10 \text{ kHz} = 20 \text{ kHz}$

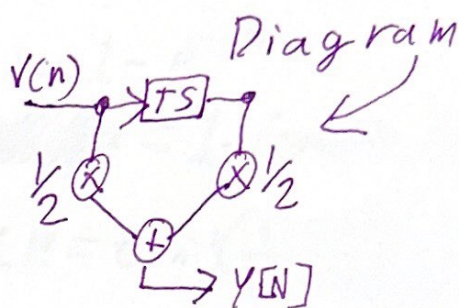
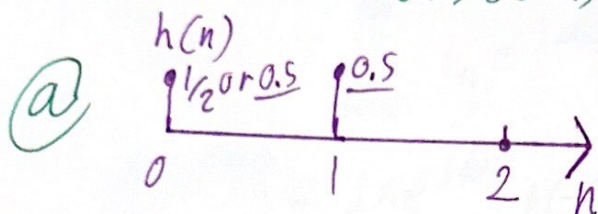
Amplitude and Phase Shift have no effect on Min. Sampling Rate.



(c) AVG Power = $\boxed{\frac{K^2}{2}}$

$$h[n] = 0.5\delta(n) + 0.5\delta(n-1)$$

Quiz 8

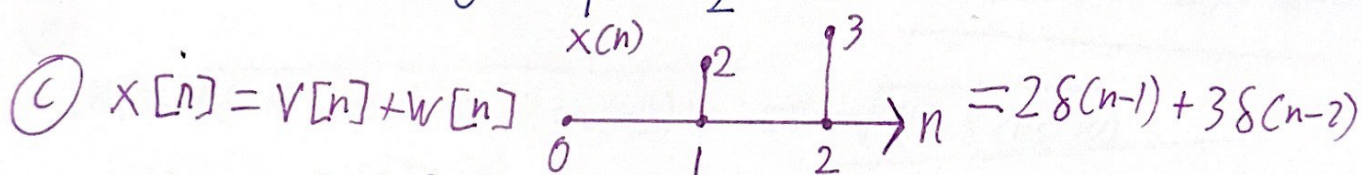
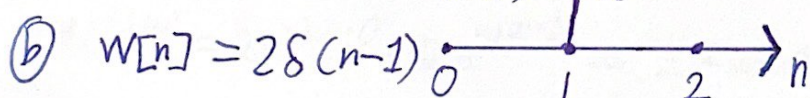
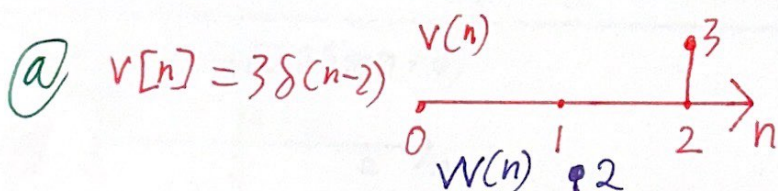


(b) $v[n] = a\delta(n) + 2b\delta(n-1)$

$$y[n] = 0.5a\delta(n) + 0.5a\delta(n-1) + b\delta(n-1) + b\delta(n-2)$$

$$y[n] = \underline{0.5a\delta(n) + (0.5a + b)\delta(n-1) + b\delta(n-2)}$$

Quiz 9



$$h[n] = \underline{\delta(n) + \delta(n-1)}$$

$$y[n] = 2\delta(n-1) + 3\delta(n-2) + 2\delta(n-2) + 3\delta(n-3)$$

$$= \underline{2\delta(n-1) + 5\delta(n-2) + 3\delta(n-3)}$$

$$V[n] = 3 \cos(0.5\pi n + \phi)$$

Note ↓

Quiz 10

$$(a) H(e^{j\omega}) = \sum_{n=0}^M b_n e^{-j\omega n} \quad e^{-j\omega} = \cos(\omega) - j \sin(\omega)$$

$$H(e^{j\omega}) = 1 \times e^{-j\omega \cdot 0} + (-1) \times e^{-j\omega \cdot 1} = 1 - e^{-j\omega}$$

$$\cancel{1 - e^{-j\omega}} = 1 - (\cos(\omega) - j \sin(\omega))$$

$$(b) |H| = \sqrt{(1 - \cos(\omega))^2 + (\sin(\omega))^2} \quad \angle H = \tan^{-1}\left(\frac{\sin(\omega)}{1 - \cos(\omega)}\right)$$

$$= \sqrt{2 - 2\cos(\omega)}$$

$$(c) |Y| = 3 \cdot |H(e^{j \cdot 0.5\pi})| \rightarrow |Y| = 3 \cdot \sqrt{2 - 2\cos(\pi/2)} = \underline{3\sqrt{2}}$$

$$\angle Y = \tan^{-1}\left(\frac{\sin(0.5\pi)}{1 - \cos(0.5\pi)}\right) = \tan^{-1}(1) = \underline{45^\circ}$$

$$V[n] = 2 \sin(0.25\pi n + \phi)$$

Quiz 11

$$(a) \begin{array}{c} h(n) \\ \uparrow \quad \uparrow \\ 0 \quad 1 \quad 2 \end{array} \xrightarrow{n}$$

$$H(e^{j\omega}) = 2e^{-j\omega \cdot 0} + 2e^{-j\omega \cdot 1} = 2 + 2e^{-j\omega} = 2 + 2\cos(\omega) - 2j\sin(\omega)$$

(b)

$$|H| = \sqrt{(2 + 2\cos(\omega))^2 + (2\sin(\omega))^2} = \sqrt{8 + 8\cos(\omega)}$$

$$\angle H = \tan^{-1}\left(\frac{2\sin(\omega)}{2 + 2\cos(\omega)}\right)$$

$$(c) |Y| = 2 \cdot \sqrt{8 + 8\cos(\pi/4)}$$

$$\angle Y = \tan^{-1}\left(\frac{2\sin(\pi/4)}{2 + 2\cos(\pi/4)}\right)$$

$$g(n) = \delta(n) + \delta(n-D)$$

Quiz 12

$$(a) \quad G(k) = \sum_{n=0}^{N-1} (\delta(n) + \delta(n-D)) e^{-j2\pi kn/N}$$

$$= \cancel{\delta(n)} e^{-j2\pi k \cdot 0/N} + \cancel{\delta(n-D)} e^{-j2\pi k \cdot D/N}$$

$$G(k) = \underline{1 + e^{-j2\pi k D/N}}$$

$$(b) \quad G(k) = 1 + e^{-j2\pi k(N-1)/N}$$

$$= 1 + e^{-j2\pi k} e^{j2\pi k/N}$$

$$= \underline{1 + e^{j2\pi k/N}}$$

$$G(0) = 1 + 1 = \underline{2}$$

$$g(n) = 2\delta(n) + \delta(n-Q) \quad \leftarrow Q$$

$$2 + e^{-j2\pi k Q/N}$$

$$G(k) = \underline{2 + e^{-j2\pi k Q/N}}$$

Quiz 13

$$(b) \quad k=0: G(0) = 2 + e^{-j2\pi \cdot 0 \cdot 2/4} = 2 + 1 = \underline{3}$$

$$k=1: G(1) = 2 + e^{-j\pi \cdot 1} = \underline{1}$$

$$k=2: G(2) = 2 + e^{-j\pi \cdot 2} = \underline{3}$$

$$k=3: G(3) = 2 + e^{-j\pi \cdot 3} = \underline{1}$$

$$g(n) = a_1 g(n-1) + a_2 g(n-2) + x(n)$$

Quiz 15

$$Z(g(n)) = G(z)$$

$$Z[g(n-1)] = z^{-1}G(z)$$

$$Z[g(n-2)] = z^{-2}G(z)$$

$$Z[x(n)] = X(z)$$

$$G(z) = a_1 z^{-1}G(z) + a_2 z^{-2}G(z) + X(z)$$

$$H(z) = \frac{G(z)}{X(z)} = \frac{1}{1 - a_1 z^{-1} - a_2 z^{-2}} \rightarrow z^2 - a_1 z - a_2$$

No Zeros Poles: $\frac{a_1 \pm \sqrt{a_1^2 - 4 \cdot 1 \cdot a_2}}{2 \cdot 1}$

$$\textcircled{b} \frac{\frac{2}{3} \pm \sqrt{\left(\frac{2}{3}\right)^2 - 4 \cdot (-\frac{1}{9})}}{2} = \frac{\frac{2}{3} \pm \sqrt{\frac{4}{9} + \frac{4}{9}}}{2}$$

$$\frac{\frac{2}{3} \pm \frac{2}{3}\sqrt{2}}{2} \rightarrow \frac{\frac{2}{3} - \frac{2}{3}\sqrt{2}}{2}, \frac{\frac{2}{3} + \frac{2}{3}\sqrt{2}}{2}$$

Magnitude Responses are equivalent due to ONLY a time shift

Quiz 16

(b) $G(z) = 2 + z^{-1}$

$$R(z) = 2z^{-2} + z^{-3}$$

$$Y(z) = 4z^{-2} + 4z^{-3} + z^{-4}$$

$$y(n) = 4\delta(n-2) + 4\delta(n-3) + \delta(n-4)$$

$$h(n) = -0.8\delta(n) + \delta(n-1)$$

$$H(k) = \sum_{n=0}^{N-1} (-0.8\delta(n) + \delta(n-1))e^{-j2\pi kn/8}$$

$$-0.8e^{-j\pi k \cdot 0} + e^{-j\pi k} = \underline{-0.8 + e^{-j\pi k}}$$

Quiz 18

@ 0 $\rightarrow -0.8 + e^{j\pi \cdot 0} = 1 - 0.8 = \boxed{0.2}$

@ 1 $\rightarrow -0.8 + e^{j\pi \cdot 1} = -0.8 - 1 = \boxed{-1.8}$

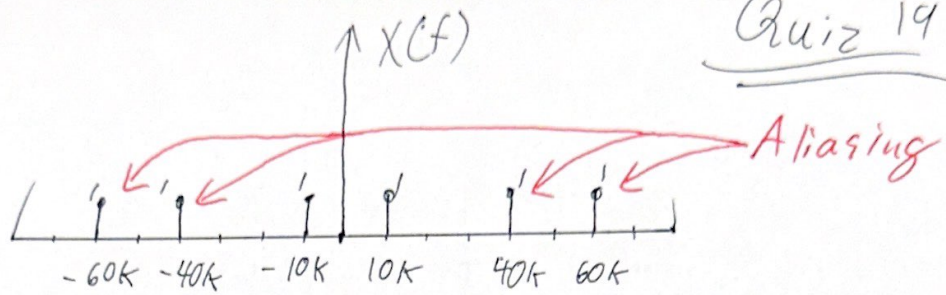
(b) $Y(z) = 0.8Y(z)z^{-1} - 0.8X(z) + X(z)z^{-1}$

$$Y(z)(1 - 0.8z^{-1}) = X(z)(-0.8 + z^{-1}) \quad 1/0.8 = 1/2$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{-0.8 + z^{-1}}{1 - 0.8z^{-1}} \Rightarrow \begin{array}{l} \text{Poles: } 1 - 0.8z^{-1} = 0, \underline{z = 0.8} \\ \text{Zeros: } z = 1/0.8 = \underline{1.25} \end{array}$$

$$H(e^{j\omega}) = \boxed{\frac{-0.8 + e^{-j\omega}}{1 - 0.8e^{-j\omega}}}$$

Quiz 19



(b) $x(n) = x(t) |_{t=nT_s}$

(c) ???